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Boosting Bass with Ported Speakers

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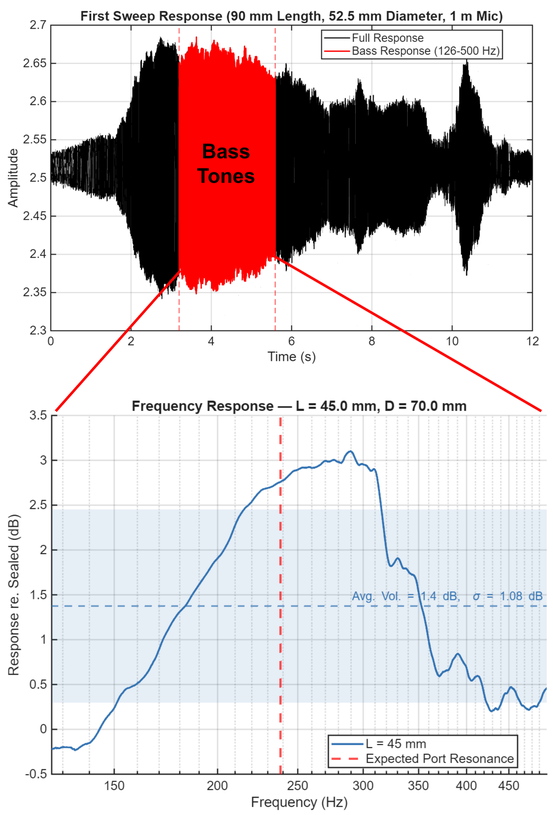
2.671 Measurement and Instrumentation

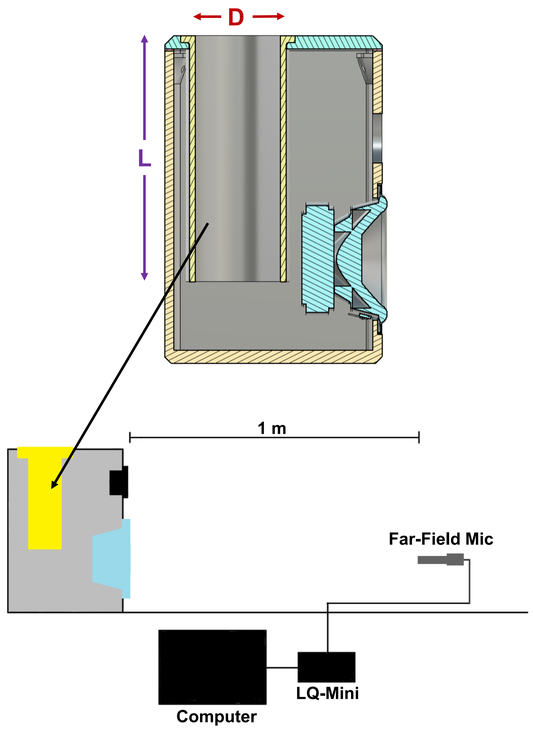
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Abstract

Ported speaker enclosures improve low-frequency output by introducing an energy-storing air mass, known as a Helmholtz resonator, but port geometry is typically selected by heuristics or lumped-parameter models that do not fully capture real acoustic behavior. This study empirically measures how port diameter and length affect the frequency response of a 3D-printed desktop bass-reflex speaker. A microphone recorded output at 1 meter in response to a 20 Hz – 20 kHz sine sweep, and average sound pressure level (SPL) and RMS deviation were evaluated from 126 Hz – 500 Hz across four diameters (17.5 – 70 mm) and five lengths (0 – 180 mm). Increasing the diameter from 17.5 mm to 70 mm increased volume by approximately 2.2 dB and decreased RMS deviation by 0.6 dB. Increasing the length from 0 mm to 180 mm decreased volume by approximately 0.9 dB and increased RMS deviation by 0.2 dB.

Experimental Design





Trial Diameters	Trial Lengths
0 mm	0 mm
17.5 mm	45 mm
35.0 mm	90 mm
52.5 mm	135 mm
70.0 mm	180 mm

$$\text{Average Volume} = \frac{1}{f_f - f_i}\int_{f_i}^{f_f} SPL(f)\,\mathrm{d}f$$

$$\text{RMS Deviation} = \sqrt{\int_{f_i}^{f_f} (SPL(f) - \text{Avg. Vol.})^2\,\mathrm{d}f}$$

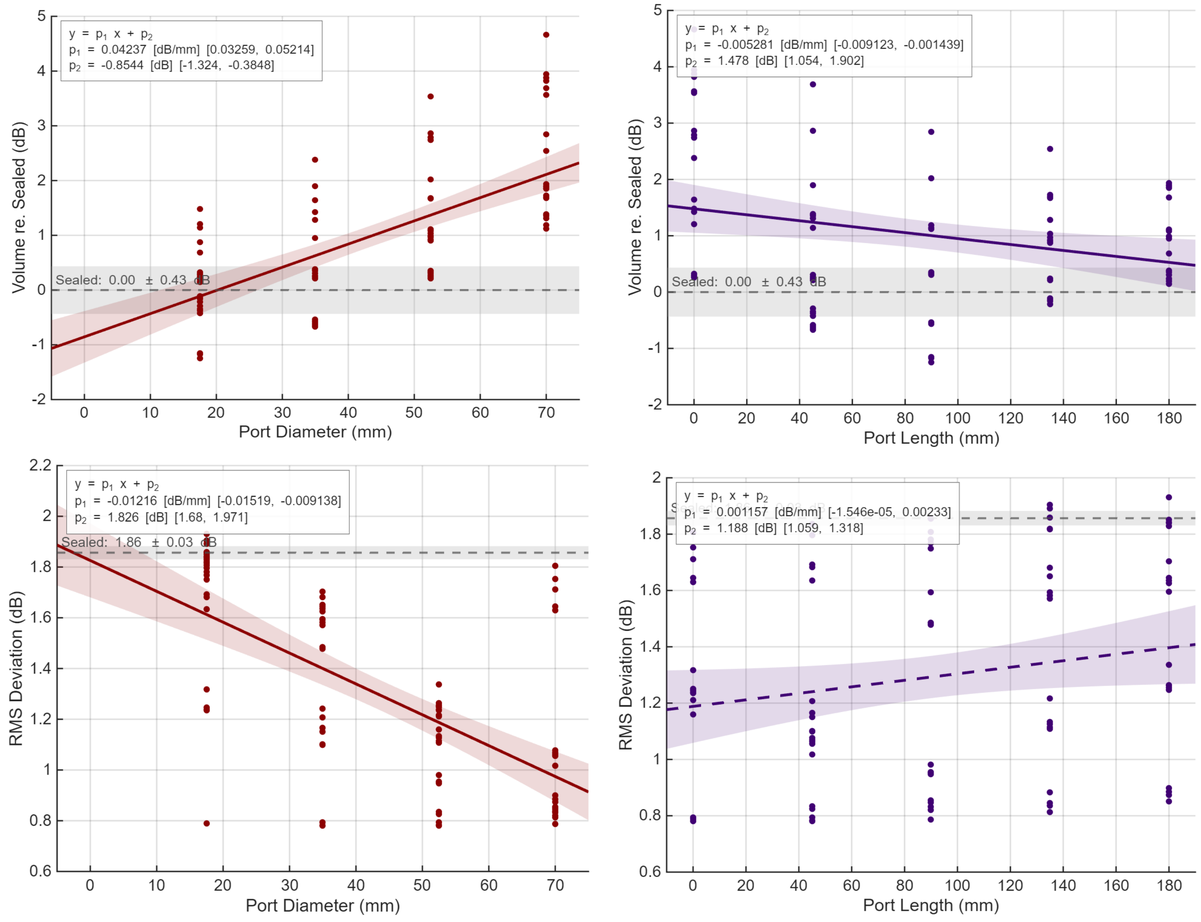
$$f_p = \frac{c}{2\pi}\sqrt{\frac{S}{VL}}$$

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Results

Plots of **Volume** and **RMS Deviation** versus **Port Length** and **Port Diameter**.



Conclusions

- Increasing port diameter significantly increased average bass output and improved response uniformity, at a rate of 0.042 dB/mm and 0.012 dB/mm, respectively.
- Increasing port length significantly decreased average bass output at a rate of 0.005 dB/mm and had no significant effect on response uniformity.
- Both trends are directionally consistent with lumped-parameter model predictions, which predict output increasing with cross-sectional area and decreasing with port-length.
- The fourth-order bass-reflex transfer function predicts Helmholtz resonance frequencies ranging from 31 – 641 Hz across the tested ports, which falls out of the tested range of 126 – 500 Hz. This likely limits the effectiveness of Helmholtz resonance theory in predicting responses according to port geometry.

References

[1] https://pngtree.com/freepng/music-audio-speaker-on-transparent-background_20343470.html

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